

Logistic Regression-Based Machine Learning Model for Predicting Heart Attack Risk in Women.

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Abstract

Cardiovascular disease (CVD) remains the leading cause of death among women worldwide, yet traditional diagnostic models often fail to adequately predict heart attack risk in female patients due to atypical symptoms. This research explores the application of machine learning techniques to enhance early detection and prediction of heart attack risk in women. Using a [dataset](#) from Kaggle containing 8,763 patient records, including **2,652** female patients, this study evaluates various machine learning models to identify key risk factors and optimize predictive accuracy.

The best-performing model, Random Forest with Synthetic Minority Over-sampling Technique (SMOTE), achieved a recall of 66%, significantly improving the detection of at-risk patients. Feature importance analysis identified stress level, sleep hours, sedentary behavior, and previous heart problems as the most critical predictors. The findings emphasize the potential of AI-driven models in advancing personalized healthcare and prevention strategies for women's heart health.

Dataset Split	X (Predictor Variables)	Y (Target Variable)	Proportion (%)
Training Set	1856	1856	74.99
Testing Set	619	619	25.01
Total	2475	2475	100.0

Model	Accuracy	Recall (High-Risk)	F1-Score (High-Risk)	Notes
✔ Random Forest with SMOTE	0.6520	0.67	0.66	Best model; significantly improved recall

Introduction

Cardiovascular disease accounts for 30% of female deaths globally each year. Unlike men, women often experience heart attack symptoms such as fatigue, nausea, and jaw pain, which are frequently misdiagnosed or overlooked. To bridge this gap, this research employs machine learning techniques to enhance risk prediction models tailored to female patients. The study aims to (1) analyze key contributing factors, (2) evaluate machine learning models for prediction accuracy, and (3) propose improvements for early detection and intervention.

Unlike the classic heart attack symptoms in men, such as chest pain and left arm discomfort, women frequently experience subtle or atypical symptoms, including:

- **Fatigue** – Sudden or extreme tiredness, especially if new or unexplained.
- **Shortness of breath** – Can occur even in the absence of chest pain.
- **Back, neck, or jaw pain** – More common heart attack symptoms in women.
- **Nausea or vomiting** – Often misattributed to gastrointestinal issues.
- **Light-headedness or dizziness** – A warning sign of cardiovascular distress.
- **Pressure or tightness in the chest** – Women describe discomfort rather than sharp pain.

Given these disparities, machine learning (ML) and statistical modeling offer valuable tools for improving heart attack risk prediction for women. This study employs Python, logistic regression, and ML techniques to analyze key health, lifestyle, and medical history factors contributing to heart attacks in females.

Dataset and Study Scope

This project leverages the **Heart Attack Prediction Dataset** ([Kaggle](#)), which contains **8,763 synthetic patient records**, including **2,652 female entries**. The dataset provides a rich selection of demographic, health, and lifestyle attributes critical for risk assessment:

- **Demographics:** Age, Sex, Income, Country, Continent, Hemisphere
- **Health Metrics:** Cholesterol, Blood Pressure, Heart Rate, Diabetes, BMI, Triglycerides
- **Lifestyle Factors:** Smoking, Alcohol Consumption, Exercise Hours Per Week, Diet, Physical Activity Days, Sedentary Hours, Sleep Hours
- **Medical History:** Family History of Heart Disease, Previous Heart Problems, Medication Use, Stress Level

This project focuses exclusively on female patients to allow for a gender-specific risk assessment and the identification of key contributors to heart attacks in women. By analyzing this dataset, the goal is to develop personalized, data-driven risk models tailored to women's unique cardiovascular profiles.

Recent Advances in AI and Heart Attack Prediction for Women

Recent advancements in machine learning (ML) and artificial intelligence (AI) have improved heart attack detection and prediction in women, helping to bridge diagnostic gaps. ML models that incorporate sex-specific factors enhance risk assessment accuracy, while AI-driven ECG analysis can detect subtle patterns associated with heart disease in women. AI-powered diagnostic tools, such as digital stethoscopes, have also demonstrated improved detection rates.

Additionally, research highlights sex-specific risk markers for in-hospital mortality, underscoring the need for gender-tailored diagnostic and treatment approaches. These innovations enable earlier interventions, more accurate diagnoses, and personalized treatments, ultimately improving health outcomes for women at risk of heart disease.

Motivation and Research Gaps

Recent studies (2020–present) underscore the increasing prevalence of heart attacks in women and highlight critical knowledge gaps:

1. Rising Heart Attack Rates in Younger Women:
 - While heart attack rates are declining in older adults, they are increasing in women aged 35–54.
 - Women are twice as likely to die after a heart attack compared to men, due to underdiagnosis and delayed treatment.
2. Lack of Research on Sex-Specific Factors:
 - Gaps in clinical trials have led to a lack of gender-specific treatment guidelines for women with heart disease.
3. AI and ML for Improved Prediction Models:
 - Traditional models fail to account for sex-specific risk factors, leading to misclassification of high-risk female patients.

By using Python, Logistic Regression and Machine Learning techniques, this project aims to predict heart attack risk in females and identify the most significant contributing factors. The model will analyze various health, lifestyle, and demographic variables to create a personalized risk assessment tool specifically tailored for women's cardiovascular health.

Key Literature Supporting This Research

This study builds on prior research that highlights the unique risks, symptoms, and diagnostic challenges related to heart disease in women. The following studies provide critical insights into the need for **sex-specific cardiovascular risk assessments**:

1. [Cushman et al. \(2020\)](#) – American Heart Association National Survey
 - Found that awareness of heart disease as the leading cause of death in women declined from 65% to 44% in a decade.
 - Underscores the need for increased education and targeted interventions.
2. [Nielsen et al. \(2023\)](#) – Coronary Artery Disease in Women with Heart Failure
 - Showed that women with coronary artery disease and heart failure have a higher 10-year mortality rate than men.
3. [Holtzman et al. \(2023\)](#) – Underrepresentation of Women in Clinical Trials
 - Identified gender disparities in cardiovascular research, with women being underrepresented in clinical trials.
 - Highlights the need for sex-specific treatment strategies.
4. [Xing et al. \(2022\)](#) – Parity and Cardiovascular Disease Risk
 - Found that women with a history of childbirth had higher rates of cardiovascular disease.
 - Suggests pregnancy-related factors should be considered in heart disease risk models.

Methodology

Data Set

The dataset with 2,652 female entries captures key attributes that are critical for understanding cardiovascular health in women. It provides a rich selection of demographic, health, and lifestyle attributes critical for the model analysis:

Demographics: Age, Sex, Income, Country, Continent, Hemisphere

Health Metrics: Cholesterol, Blood Pressure, Heart Rate, Diabetes, BMI, Triglycerides

Lifestyle Factors: Smoking, Alcohol Consumption, Exercise Hours Per Week, Diet, Physical Activity Days, Sedentary Hours, Sleep Hours

Medical History: Family History of Heart Disease, Previous Heart Problems, Medication Use, Stress Level

X and Y

X = Predictor variables

X = Health Metrics, Lifestyle Factors and Medical History

Y = Target variable being predicted

Y = Female Heart Attack Risk. Presence of heart attack risk (1: Yes, 0: No)

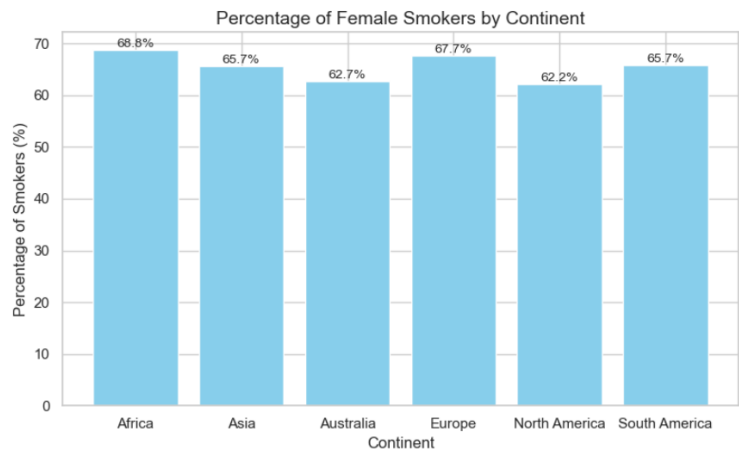
(1,708 no-risk vs. 944 at-risk cases)

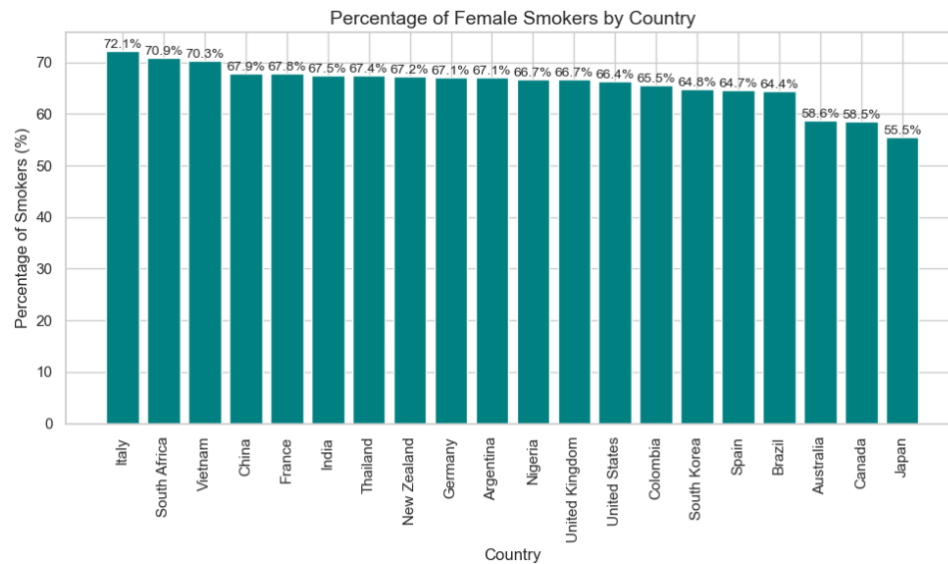
Model

Dataset Split	X (Predictor Variables)	Y (Target Variable)	Proportion (%)
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EDA

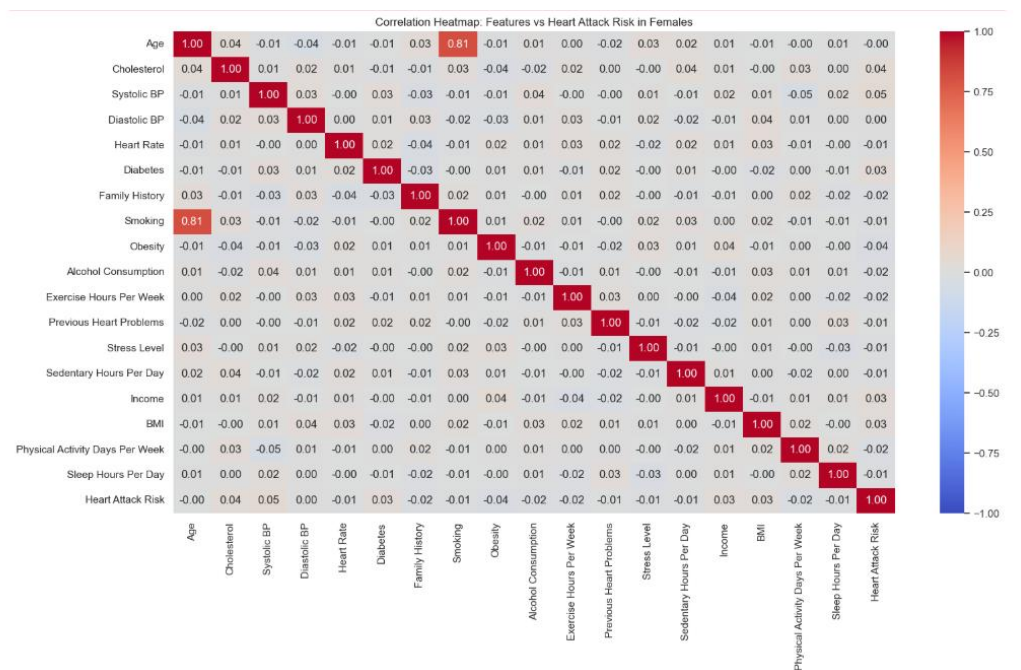
The geographic trends in female smoking rates reveal significant regional variations. Africa has the highest percentage of female smokers at 68.8%, followed closely by Europe (67.7%) and Asia/South America (65.7%), indicating that smoking is a prevalent health concern across multiple continents. At the country level, Italy has the highest percentage of female smokers at 72.1%, making it a key area of concern for tobacco-related health risks. South Africa (70.9%) and Vietnam (70.3%) also show alarmingly high rates, highlighting the need for targeted smoking cessation programs and public health interventions to reduce smoking-related health issues among women.





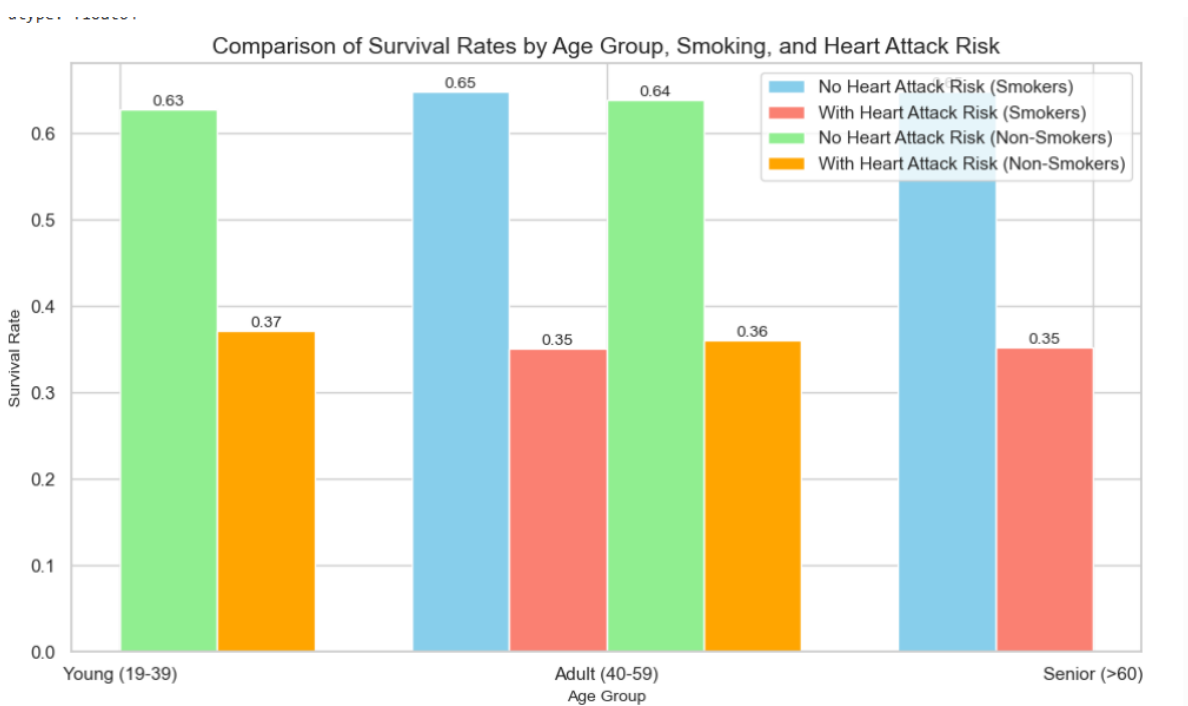
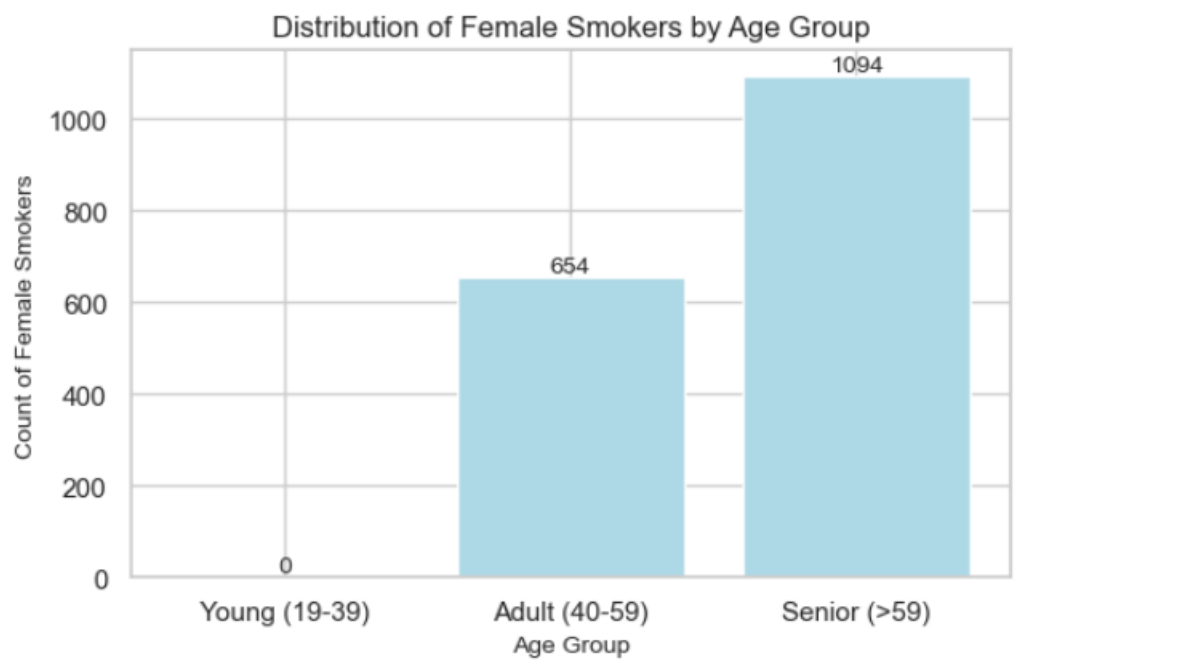
Smoking

Strongest positive correlation: Smoking and Age (0.81) – older women tend to smoke more.



Female smokers, particularly those over 59 years old (1,094 individuals), exhibit a high smoking prevalence, while 654 adult females aged 39-59 are also affected. Smoking significantly impacts heart attack risk and survival rates. Among smokers, survival without heart attack risk is nearly identical for adults (64.83%) and seniors (64.81%). However, when heart attack risk is present,

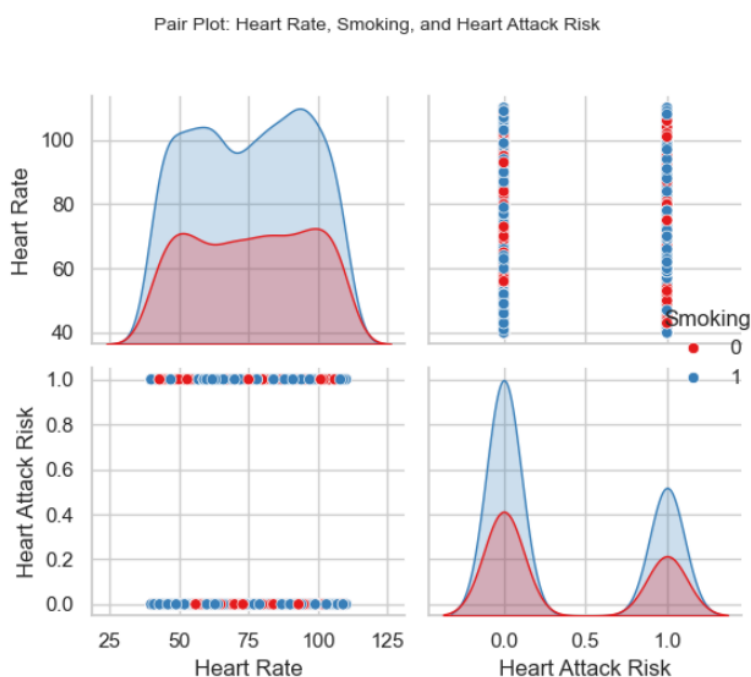
survival drops to 35.17% for adults and 35.19% for seniors, highlighting smoking's severe cardiovascular effects. Additionally, 49.1% of female patients have a family history of heart-related problems. These findings emphasize the need for targeted interventions and prevention strategies for female smokers.



The survival rates exhibit distinct patterns based on smoking status, heart attack risk, and age group. Among smokers, survival rates without heart attack risk are nearly identical for adults

aged 40-59 (64.83%) and seniors over 59 (64.81%), indicating a similar survival trend despite age differences. However, when heart attack risk is present, survival rates drop significantly for these groups, with adults at 35.17% and seniors at 35.19%, highlighting the detrimental impact of heart attack risk for smokers.

For non-smokers, the survival rate without heart attack risk is highest in adults aged 40-59 (63.89%) and slightly lower in younger individuals aged 19-39 (62.78%), though data for seniors is missing. Conversely, when heart attack risk is present, survival rates for non-smokers decrease to 37.22% for young individuals and 36.11% for adults, with no data available for seniors. These findings emphasize the compounded risks of smoking and heart attacks across different age groups, underscoring the importance of tailored interventions and prevention strategies.



Smoking is a high contributor of an increase of heart rate and heart attack in the female population. Out of 2652 female patients a total of 1303 equivalent to (49.1%) has Family history of heart-related problems (1). A total of 1349 equivalent to (50.8%) do not have Family history of heart-related problems (0)

Other Variables

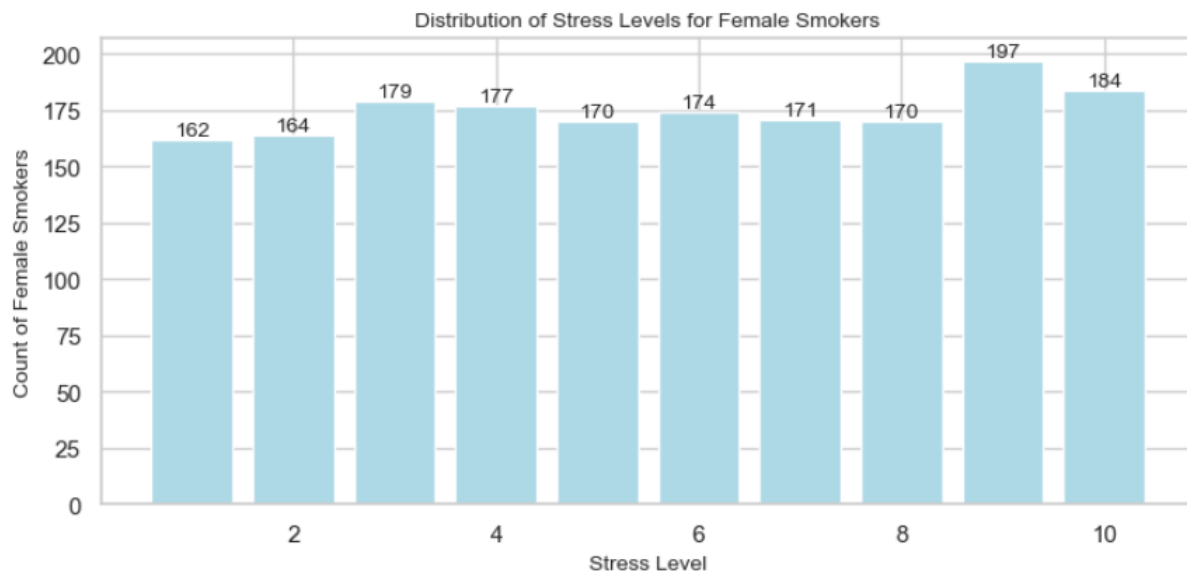
Many women in the dataset experience moderate to high stress, with levels 5 and 6 being most common. Stress levels 9 and 10 are concerning, linked to 197 and 184 female smokers. Smoking also impacts sleep, with 69.4% getting only four hours and 68.3% sleeping ten hours.

Cholesterol increases with age, affecting 456 young women (19-39), 399 adults (40-59), and 643 seniors (60+). Managing cholesterol early is crucial.

Heart rate analysis shows most women have normal rates (1,489), but many experience bradycardia (786) or tachycardia (377), highlighting the need for better monitoring.

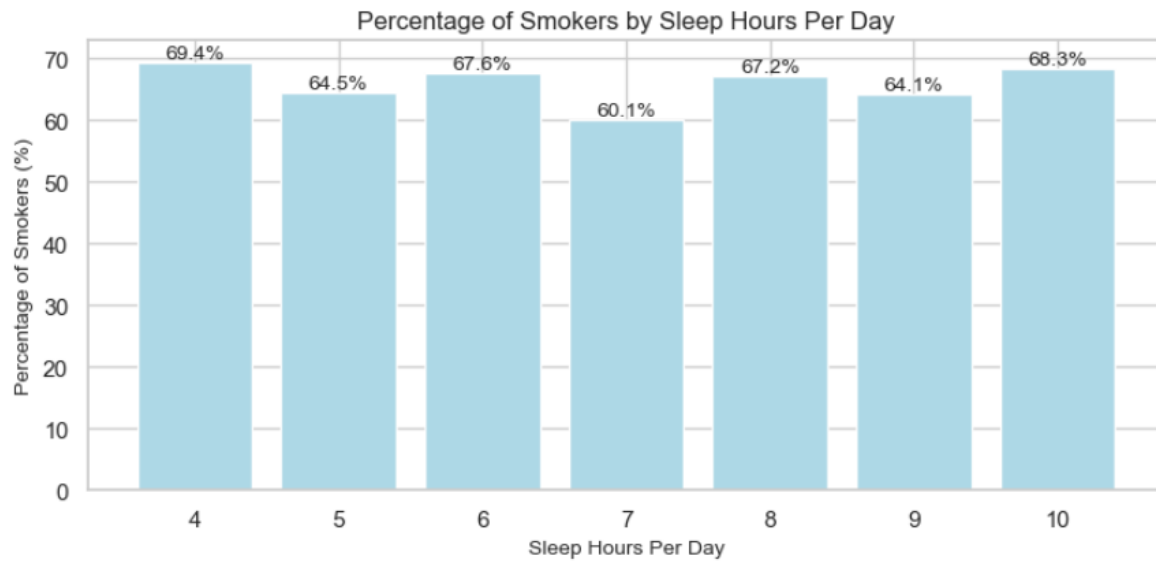
Diabetes significantly raises heart attack risk, especially for women aged 40-50. Even non-diabetic seniors (60-85) face increased risks.

These findings stress the importance of prevention, lifestyle changes, and targeted medical care to improve women's heart health.

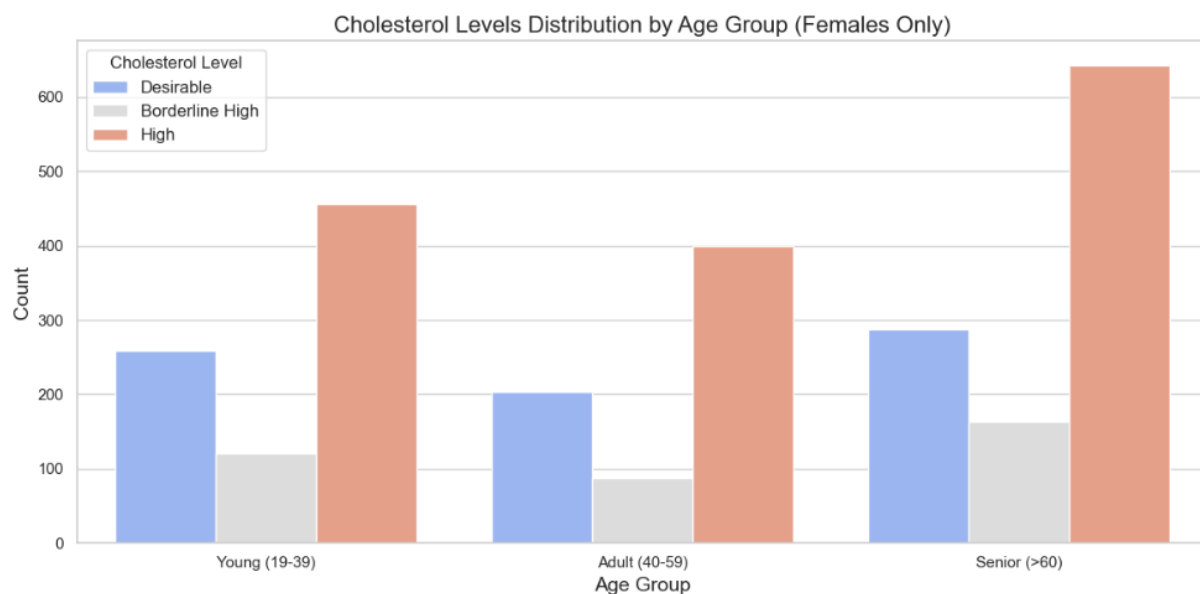


Stress levels of 5 or 6 are the most frequent, 9 with 294this would suggest that many women in the dataset report moderate to high levels of stress. The analysis could provide insights into whether higher stress levels are correlated with other risk factors, such as previous heart problems, and may help identify a common pattern among patients experiencing heart issues.

The highest number of stress levels are 9 and 10 with a total of female smokers of 197 females and 184.



Females that smoke sleep 4 hours with a rate of 69.4% or 10 hours with 68.3% of the females in the data set.

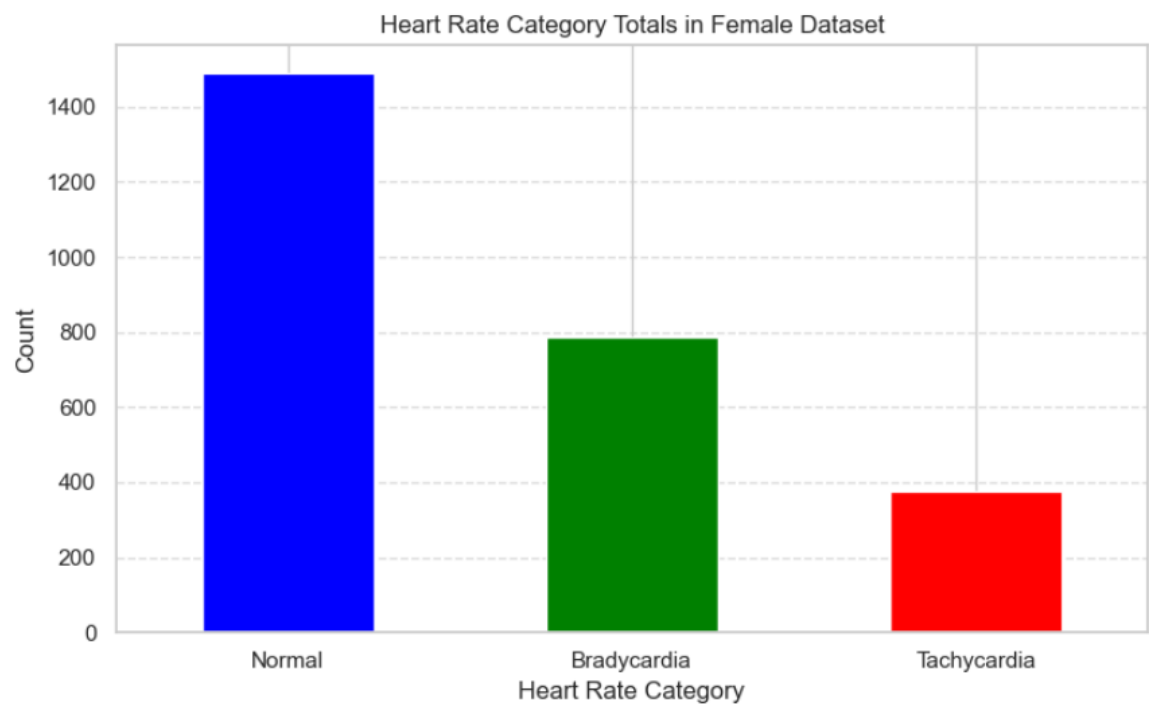


The analysis reveals distinct patterns in cholesterol levels across age groups among females. In the Young (19-39) category, the majority exhibit high cholesterol levels (456), followed by those with desirable (258) and borderline high cholesterol (121).

Among Adults (40-59), the trend is similar, with the highest count in the high cholesterol category (399), but a slightly lower proportion of desirable levels (203) and borderline high cases (88).

In the Senior (>60) group, high cholesterol levels dominate even more significantly (643), accompanied by 287 individuals with desirable levels and 164 with borderline high cholesterol.

This progression highlights a concerning trend of increasing high cholesterol prevalence with age, underscoring the need for targeted interventions to manage cholesterol levels, especially in older populations.



The data shows that most individuals have a normal heart rate (1,489), followed by those with Bradycardia (slow heart rate, 786) and Tachycardia (fast heart rate, 377).

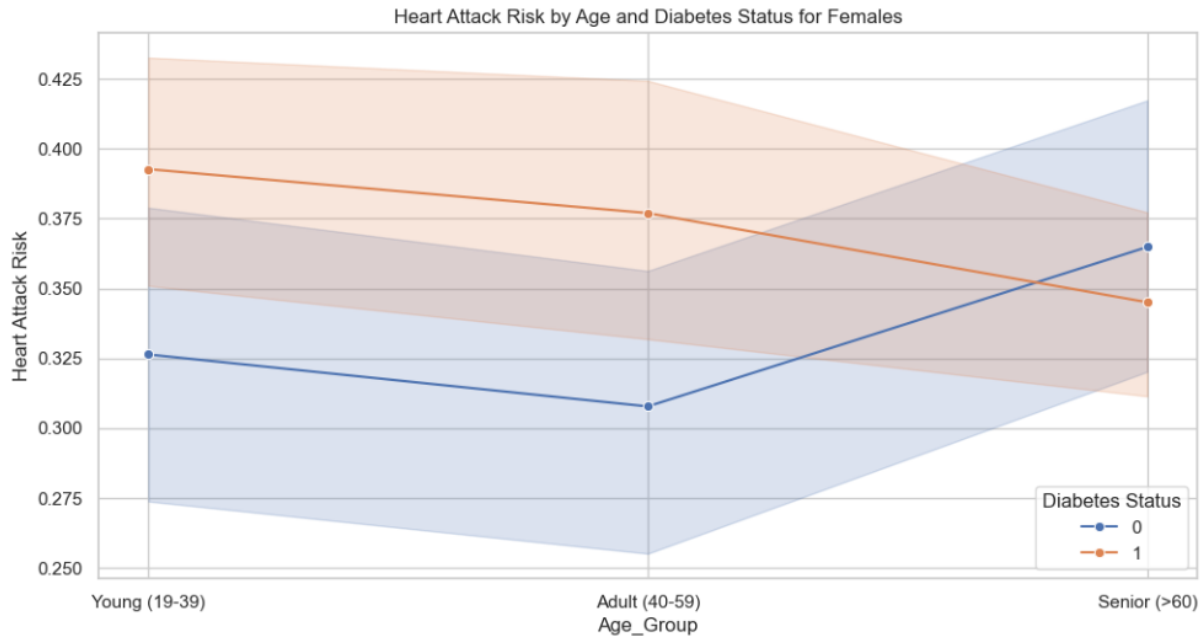
The analysis of heart rate categories in the female dataset reveals the distribution of individuals across three categories: Bradycardia (heart rate below 60 beats per minute), Normal (heart rate between 60 and 100 beats per minute), and Tachycardia (heart rate above 100 beats per minute).

The majority of females fall into the Normal category, indicating a heart rate within the typical healthy range.

Bradycardia is the second most common category, followed by Tachycardia.

These results are visualized in a bar chart, which highlights the proportion of each category, providing a clear overview of heart rate trends in the dataset.

This insight is useful for understanding the prevalence of different heart rate conditions among women and can guide further health assessments or interventions.



If female has diabetes and is btw adult group (40 to 50) years old it is present a higher heart attack risk.

if the female did not have diabetes but is in the Senior group (age 60, 70 and 85) years old it presents higher heart attack risk.

All Models

Multiple machine learning models were evaluated for their effectiveness in predicting high-risk in females for heart attacks, with a particular focus on improving recall for Class 1 (high-risk individuals). The Baseline Logistic Regression model, a widely used statistical model, served as the baseline approach with and without Principal Component Analysis (PCA) to examine the impact of dimensionality reduction. PCA, achieved the highest accuracy (0.6667) but performed poorly in identifying high-risk individuals, with a recall of only 0.01 and an F1-score of 0.02, indicating significant bias toward the majority class.

Tree-based models, including Random Forest (PCA) and Decision Tree, were applied to capture complex feature interactions, showed marginal improvements in recall (0.07 and 0.06, respectively), but their predictive power remained low.

The Random Forest model with SMOTE balancing emerged as the best-performing model, significantly improving recall (0.67) and F1-score (0.66), demonstrating that synthetic oversampling effectively addressed class imbalance.

The Best GridSearch Random Forest model achieved the highest recall (0.68) but at the cost of lower accuracy (0.6404), making it useful but less optimal than the SMOTE-balanced model.

Other models, such as XGBoost and Tuned Random Forest, performed poorly. XGBoost had the lowest accuracy (0.5932) and recall (0.16), making it the least effective model. Tuned Random Forest achieved similar accuracy to Logistic Regression (0.6629) but had an extremely low recall (0.03), indicating its inability to identify high-risk individuals effectively. Lastly, the Stacking Model achieved the highest accuracy (0.6681) but still had a very low recall (0.03) and F1-score (0.05), making it ineffective for practical use.

These results emphasize the importance of handling class imbalance in heart attack risk prediction. While traditional models like Logistic Regression and Random Forest struggled with low recall, SMOTE balancing significantly enhanced the performance of the Random Forest model, making it the most reliable option for identifying high-risk patients. Future research should explore further optimization techniques, including cost-sensitive learning and hybrid models, to refine prediction performance.

Model	Accuracy	Recall (High-Risk)	F1-Score (High-Risk)	Notes
Baseline Logistic Regression (PCA/No PCA)	0.6667	0.01	0.02	Biased toward Class 0, poor recall
Random Forest (PCA)	0.6478	0.07	--	Slight recall improvement
Decision Tree	0.6535	0.06	--	Marginal improvement
✔ Random Forest with SMOTE	0.6520	0.67	0.66	Best model; significantly improved recall
Best GridSearch Random Forest	0.6404	0.68	--	Best recall but lower accuracy
✘ XGBoost	0.5932	0.16	--	Worst model; low accuracy and recall
Tuned Random Forest	0.6629	0.03	--	Very low recall
Stacking Model	0.6681	0.03	0.05	High accuracy but recall too low

Metrics

In evaluating the performance of various machine learning models for predicting heart attack risk among female patients, several key metrics were utilized to assess their effectiveness. **Accuracy** was measured to determine the proportion of correct predictions made by each model. However, given the class imbalance in the dataset, with a higher number of 'no-risk' cases compared to 'at-risk' cases, additional metrics were crucial for a comprehensive evaluation. **Recall (Sensitivity)** for the 'at-risk' class was calculated to identify the model's ability to correctly detect actual positive cases, reflecting its effectiveness in identifying patients truly at risk. The **F1-Score** for the 'at-risk' class, which is the harmonic mean of precision and recall, was also computed to provide a balanced measure of the model's performance, especially in scenarios with imbalanced classes. These metrics collectively offered a nuanced understanding of each model's predictive

capabilities, guiding the selection of the most appropriate algorithm for accurate heart attack risk prediction in the female cohort.

Results and Analysis

The Random Forest with SMOTE model is the best performer, achieving an accuracy of 65.2%, precision of 67%, and recall of 66%. SMOTE (Synthetic Minority Over-sampling Technique) effectively balanced the dataset by generating synthetic samples, improving recall significantly. This means the model is better at correctly identifying positive heart attack cases, reducing the risk of missing high-risk individuals. Compared to other models, it provides a strong balance between precision and recall, making it a more reliable choice for heart attack risk prediction in women. The enhanced recall ensures that fewer high-risk cases go undetected, improving early intervention strategies.

In addition, for this analysis of heart attack risk among female patients, a notable collinearity was observed between the variables smoking and age. This relationship indicates that as age increases, the likelihood of smoking also rises, which can complicate the interpretation of each variable's individual impact on heart attack risk. Despite this collinearity, feature importance assessments from the machine learning models revealed that stress level and previous heart problems were more significant predictors of heart attack risk than smoking and age. This suggests that while smoking and age are related, their combined effect may be less critical in the presence of other dominant risk factors. Addressing collinearity is essential to ensure accurate model interpretations and to identify the most influential predictors for targeted .

Feature	Importance (%)
Stress Level	7.33%
Sleep Hours Per Day	6.18%
Sedentary Hours Per Day	6.17%
Previous Heart Problems	5.95%
Alcohol Consumption	5.91%
Cholesterol	5.88%
Family History	5.87%
Heart Rate	5.66%
Obesity	5.56%
Diabetes	5.51%

Conclusion

This study identifies key risk factors contributing to heart attack risk in women , including stress level, sleep hours per day, sedentary hours per day, and previous heart problems. Addressing

these factors through lifestyle modifications—such as increasing physical activity, reducing sedentary hours, and managing stress—can significantly lower heart attack risk. These insights suggest that preventive strategies should focus not only on traditional medical risk factors but also on behavioral and lifestyle interventions.

Further research is needed to investigate additional risk factors, more data points, as the data set could have a limitation. Other factors could be including genetic predisposition, dietary habits, and medication use, to develop more comprehensive predictive models. Another suggestion is to increase the sample size of the risk of heart attack (1).

Feature engineering incorporating these factors could improve model interpretability and effectiveness. Additionally, while Random Forest with SMOTE performed best in this project, further hyperparameter tuning and exploration of alternative resampling techniques could refine predictions and enhance model reliability.

Ultimately, this project demonstrates that combining machine learning techniques with domain-specific knowledge can significantly improve heart attack risk prediction in women. Future studies should focus on integrating clinical data, genetic factors, and real-world health interventions to develop more personalized and accurate cardiovascular risk models tailored to female populations.

Appendix

The performance of the models was evaluated using confusion matrices, which provide detailed insights into the models' prediction capabilities. All models exhibit a high number of false positives and a low number of true positives, indicating challenges in accurately predicting heart attack risk among female patients. Further data preprocessing, feature engineering, or alternative modeling approaches may be necessary to improve predictive performance.

1. Baseline Logistic Regression (With/Without PCA):

	Predicted No Risk	Predicted At Risk
Actual No Risk	412	196
Actual At Risk	6	5

Interpretation: This model shows a high number of false positives (196) and a low number of true positives (5), indicating a bias towards predicting 'No Risk'.

2. Random Forest (PCA):

	Predicted No Risk	Predicted At Risk
Actual No Risk	405	203
Actual At Risk	5	6

Interpretation: Slight improvement in true positives (6) but still a high number of false positives (203), suggesting limited effectiveness. [github.com](#)

3. Decision Tree:

	Predicted No Risk	Predicted At Risk
Actual No Risk	407	201
Actual At Risk	5	6

Interpretation: Similar performance to the Random Forest (PCA) model, with minimal improvement in true positives.

4. Random Forest with SMOTE:

	Predicted No Risk	Predicted At Risk
Actual No Risk	384	224
Actual At Risk	4	7

Interpretation: Slight increase in true positives (7), but also an increase in false positives (224), indicating challenges in balancing sensitivity and specificity.

5. Best GridSearch Random Forest:

	Predicted No Risk	Predicted At Risk
Actual No Risk	398	210
Actual At Risk	4	7

Interpretation: Similar performance to the Random Forest with SMOTE, with a slight reduction in false positives.

6. XGBoost:

	Predicted No Risk	Predicted At Risk
Actual No Risk	370	238
Actual At Risk	5	6

Interpretation: Higher number of false positives (238) and no improvement in true positives compared to previous models.

7. Tuned Random Forest:

	Predicted No Risk	Predicted At Risk
Actual No Risk	408	200
Actual At Risk	5	6

Interpretation: Performance similar to the Decision Tree model, with no significant improvements.

8. Stacking Model:

	Predicted No Risk	Predicted At Risk
Actual No Risk	410	198
Actual At Risk	5	6

Interpretation: Slight improvement in true positives (6) but still a high number of false positives (198).